

Client/Institute Name	:	Expleo/KIOT
Course	:	SDET
Technical Assessment	:	25
Module	:	TypeScript and Playwright Fundamentals
Date	:	01-Jul-26
Time	:	02:00 P.M – 05:00 P.M
Duration	:	3 h
Total Marks	:	40 Marks

Part A – 1 x 10 = 10 Marks

Part B – Linux Commands (30 Marks)

1. Write a typescript that reads two integers from the user, verifies if both inputs are valid integers, performs addition, subtraction, multiplication, and division, and prints the results. Handle invalid inputs gracefully. (05 Marks)
2. Write typescript for the given scenario using function: Reina and Sierra were playing a game. Reina would give out number, and Sierra must reverse the given number. Help Sierra by writing program to reverse the number. Note that Reina should give a five-digit number. (05 Marks)
3. Create playwright Firefox tests in headed mode for the following: (05 Marks)
 - Launch "https://www.bing.com/"
 - Get title
 - Get current URL
 - Verify Title
 - Verify URL.
 - Get inner text of any element
 - Get some attributes of any element.
 - Locate search text field and type playwright and click the first link and assert the title.

4. Design test scripts and framework components for the following requirements: (25 Marks)

Test Case 01: Validate New User Registration

Launch the browser and navigate to <https://tutorialsninja.com/demo>

Navigate to the Register page from the My Account menu.

Fill in the registration form with valid, dynamically generated user details and submit it.

Expected Result:

A confirmation message,

"Your Account Has Been Created!"

is displayed.

Test Case 02: Validate Login with Multiple Data Sets

Navigate to the Login page.

Using Playwright Data-Driven Testing (test.describe() with multiple test data or looping through datasets), execute the login flow with at least three data sets:

Valid credentials

Invalid email format

Incorrect password

Capture and validate the resulting page or error message for each data set.

Expected Result:

Valid credentials redirect to the My Account page.

Invalid email format displays the appropriate validation or login error.

Incorrect password displays the corresponding error message.

Test Case 03: Validate Product Search Functionality

On the Home page, search for a product (for example, MacBook) using the search box.

Capture the list of products displayed on the Search Results page.

Validate that the results contain the expected product.

Expected Result:

The Search Results page displays at least one product containing the searched keyword.

Requirements to be Met

Design Page Object Model (POM) classes for all pages involved:

Home Page

Registration Page

Login Page

Search Results Page

Product Page

Optimize the POM framework by implementing reusable/common methods such as:

Navigation methods

Waiting utilities

Click helper methods

Text retrieval methods

Element visibility checks

Group the test cases using Playwright Test Groups and Tags:

Smoke Suite

Test Case 01

Test Case 03

Regression Suite

Test Case 02

Implement Hooks or Fixtures for common browser setup and teardown across all test cases.

Demonstrate execution of:

An individual test case

Tests grouped by tags

Parallel execution using Playwright Test Runner

Every test case must include appropriate Playwright Assertions (expect) to validate the expected outcome.

Generate and view the Playwright HTML Test Report after execution.